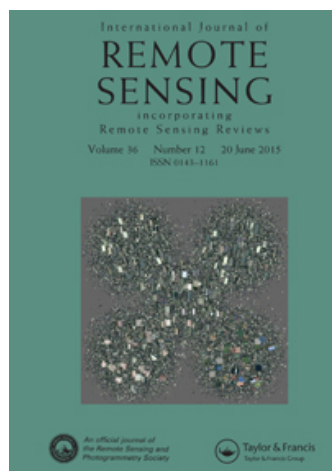


This article was downloaded by: [University of Otago]

On: 30 July 2015, At: 00:16

Publisher: Taylor & Francis

Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: 5 Howick Place, London, SW1P 1WG



[Click for updates](#)

International Journal of Remote Sensing

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/tres20>

Road network extraction: a neural-dynamic framework based on deep learning and a finite state machine

Jun Wang^a, Jingwei Song^a, Mingquan Chen^b & Zhi Yang^a

^a Institute of Remote Sensing and Digital Earth Chinese Academy of Sciences, University of Chinese Academy of Sciences, Beijing, China

^b Department of Geography, University of California, Santa Barbara, Santa Barbara, CA, USA

Published online: 30 Jun 2015.

To cite this article: Jun Wang, Jingwei Song, Mingquan Chen & Zhi Yang (2015) Road network extraction: a neural-dynamic framework based on deep learning and a finite state machine, International Journal of Remote Sensing, 36:12, 3144-3169, DOI: [10.1080/01431161.2015.1054049](https://doi.org/10.1080/01431161.2015.1054049)

To link to this article: <http://dx.doi.org/10.1080/01431161.2015.1054049>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms &

Road network extraction: a neural-dynamic framework based on deep learning and a finite state machine

Jun Wang^{a*}, Jingwei Song^a, Mingquan Chen^b, and Zhi Yang^a

^a*Institute of Remote Sensing and Digital Earth Chinese Academy of Sciences, University of Chinese Academy of Sciences, Beijing, China;* ^b*Department of Geography, University of California, Santa Barbara, Santa Barbara, CA, USA*

(Received 27 December 2014; accepted 4 April 2015)

Extracting road networks from very-high-resolution (VHR) aerial and satellite imagery has been a long-standing problem. In this article, a neural-dynamic tracking framework is proposed to extract road networks based on deep convolutional neural networks (DNN) and a finite state machine (FSM). Inspired by autonomous mobile systems, the authors train a DNN to recognize the pattern of input data, which is an image patch extracted in a detection window centred at the current location of the tracker. The pattern is predefined according to the environment and associated with the states in the FSM. A vector-guided sampling method is proposed to generate the training data set for the DNN, which extracts massive image-direction pairs from the imagery and existing vector road maps. In the tracking procedure, the size of the detection window is determined by a fusion strategy and the extracted image patches represent the orientation features of the road (local environment) that can be recognized by the trained DNN. The reactive unit in FSM associates states with behaviours of the tracker while continually modifying the orientation to follow the road and generating a sequence of states and locations. In this way, our framework combines the DNN and FSM. DNN acts as a key component to recognize patterns from a complex and changing environment; FSM translates the recognized patterns to states and controls the behaviour of the tracker. The results illustrate that our approach is more accurate and efficient than the traditional ones.

1. Introduction

Very-high-resolution (VHR) imagery is an important data source for information updates in geographical information system (GIS). Timely road map updates are crucial in applications that include disaster management, urban planning, and impervious surface extraction (Weng 2012; Peng et al. 2008; Singh and Garg 2014). The distribution of roads is also a crucial factor in recent scene understanding research (Gerl, Kreibich, and Bochow 2003). However, traditional manually performed road extraction is time consuming and road-network data in GIS systems are usually out of date or inaccurate because of mistakes in human operations. In the last three decades, there has been extensive research on reducing or replacing the work human involved in a robust way. VHR satellite and aerial imagery has provided not only detailed road information but also obstructions/interference, e.g. shadows, vehicles, and trees along the road. Extracting roads from such imagery remains a relevant and challenging topic. The major problems are the following:

*Corresponding author. Email: wangjun@radi.ac.cn

(1) Insufficient context problems. In satellite imagery, say QuickBird imagery, a pixel can comprise a square piece of land with a width of 0.6 m. Given the complexity in recognizing roads, the typical detecting windows contain local intensity information and are too small (e.g. 4, 8, or 16 pixels) to incorporate enough context (Huang and Zhang 2013). Moreover, in aerial imagery with a resolution of 0.2 m/pixel, road features cannot be fully captured within the typical detecting windows. It is arduous to identify whether the objects captured are road segments. As Figure 1(a) shows, the object may be predicted to be a road segment in terms of radiation and spectral information; however, it is actually cropped from the roof of the apartment building. Figure 1(b) shows the roof of an apartment building with larger context.

(2) Poorly structured problems. Road details can be observed through VHR imagery; however, disturbance and errors are also included. As shown in Figure 1(c), the structure of the road segment is irregular and hard to be modelled. Figure 1(d) shows a road segment with trees on both sides. The disturbances vary from intensity and structure, which makes it difficult for the algorithm to extract and then recognize road features.

Many previous studies have presented road extraction methods that can be categorized as either *ad hoc* or learning-based methods.

(1) *Ad hoc* methods.

If the individual road features in images can be featured as having (a) a relatively constant width, (b) a uniform surface intensity, or (c) a path that is locally well fitted using a parabola (McKeown, David, and Denlinger 1988; Da Silva and Centeno 2012). Then, *ad-hoc* methods can be applied to focus on using these features and attempt to follow predefined modelling methodologies (Liu et al. 2013; Wang and Zhang 2011).

A fast linear feature detector for road extraction was proposed by Shao et al. (2011). Moreover, Xiangyun and Tao (2007) exploited the common experience that salient linear features are always perceived globally with little reasoning or high-level knowledge involved. Centre points and segments detected at the low-resolution image level were hierarchically grouped and then analysed with topology constrains. A similar technique that introduced multiscale geometric analysis was utilized to extract roads (He et al. 2012).

The Radon transform is a linear feature detector and possesses the ability to transform images to a domain of possible line parameters. Bright lines in images will give a peak in the visualization of the corresponding parameters. Zhang and Couloigner (2007) investigated the peak selection problem in the line detector and proposed a mean filter to locate the true peak in the Radon image. Li et al. (2010) proposed the revised parallel-beam

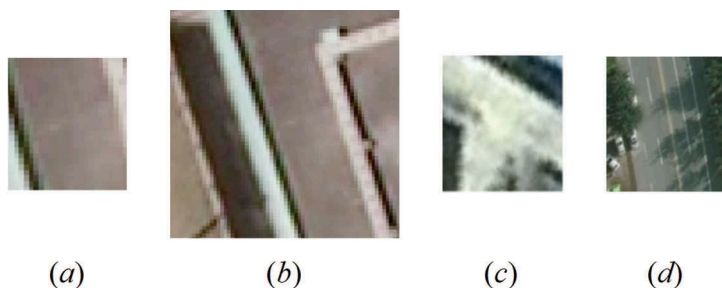


Figure 1. Insufficient context and poorly structured problems. (a) Objects cropped in the typical detection window. (b) The roof of the building with larger context. (c) An ill-structured road segment. (d) The disturbances of trees on both sides of roads.

Radon transform by solving the problem of the false maximum of the coefficients in the Radon domain. Although straight road edges can be extracted successfully by these algorithms; there are still difficulties in processing complicated images.

Template-matching based approaches are often used in extracting objects in imagery (Zhang et al. 2011). Lin et al. (2011) proved that line-shaped lane markings or median strips are less impacted by disturbances of vehicles or shadows of trees than any other parts in imagery. A least-squares template matching is utilized to search the centre line points. A double window model that consists of the outer fixed square window and the inner rotary rectangular one is set in the local detection step (Cheng et al. 2012). The outer window is used to obtain the local road direction, and the inner one searches the centre points of a road segment. Template-matching based approaches are simple and efficient in extracting roads but may not be robust enough to recognize roads with great variations in VHR imagery.

Road networks consist of road junctions and straight/curved paths (Miao et al. 2014). Many methods attempted to model the road junctions and improve the results of network extraction. Hu et al. (2007) defined a spoke wheel operator that detects the footprints of pixels. The proposed toe-finding algorithm was designed to detect the dominant directions of the footprints. However, it may suffer from over extraction and it is very hard to remove non-road nodes in pruning step.

The road network is also a kind of line-network, which can depict the centre lines (Montoya-Zegarra et al. 2014). Chai, Forstner, and Lafarge (2013) provided a structurally coherent solution based on stochastic geometry and the Monte Carlo mechanism. The salient aspects of the method are accurate and flexible. However, the sampling procedure of junction-point processes may be time consuming, as remote-sensing images in real applications are much larger than retinal images or blood vessel images.

(2) Learning-based methods.

Road extraction from imagery is actually a classification problem (Mnih 2013; Singh and Garg 2014). Features can be extracted and trained by many machine learning algorithms, e.g. neural networks and support vector machine (SVM) (Huang and Zhang 2009). However, due to lack of labelled samples and the complexity of road features, it is hard to make the most of the power in modern classifiers.

Barsi and Heipke (2003) proposed a road junction detector by training a neural network on a training data set of 542 junction and non-junction samples. Spectral and structural features at different scales are extracted and trained in several SVMs (Huang and Zhang 2009). The results of spectral and structural classification are integrated by applying a fusion method of majority voting. Principal component analysis (PCA) is applied to reduce the dimensionality of the input image patch, in this way large context can be used in training (Mnih and Hinton 2010; Mnih 2013). An unsupervised learning procedure was utilized for pretraining, before the restricted Boltzmann machines were trained with complicate large challenging urban data sets. These methods that learn to predict whether the objects are roads prove that it is very arduous because of the complexity of the surface.

Considering these deficiencies, the authors propose a semi-automatic neural-dynamic tracking framework based on deep convolutional neural networks (DNNs) and finite state machine (FSM). The framework consists of two processing steps: the training step and the tracking step.

2. Methodology

The proposed framework uses DNN for road pattern recognition and FSM for tracking with focus on accuracy and generalization. Figure 2 shows the flow chart of the

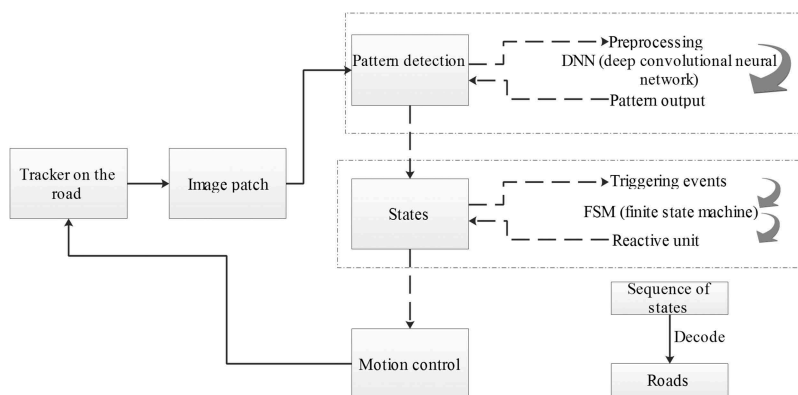


Figure 2. Overview of the framework.

framework. The intrinsic details of how each part is implemented are presented next in this section.

2.1. Problem formulation

Previous learning-based methods are generally trained by expert-labelled data and attempt to predict whether a given pixel, an image patch, or image segment (in object-based image analysis) is a road (Song and Civco 2004; Huang, Lu, and Zhang 2014; Mnih 2013). This formulation can be simplified as a function $p(M(i, j)|I)$, where I is the set of image patches and $M(i, j)$ are the corresponding labels (ground truth map) associated with I . In training data set generation, $M(i, j)$ is defined to be 1 whenever image patch I belongs to a road segment and 0 if I is non-road. The goal of these methods is to learn $p(M(i, j)|I)$ from the training data set. Many studies prove that it is arduous because of the complexity of the road surfaces.

Considering that roads are connected as a whole network, which is a unique feature on the imagery, we implement the road extraction by tracking from an initial point on the road. The key problem of tracking is how to adjust the turning angle φ of the tracker based on the current environment automatically and precisely. Suppose that the tracker is on the current location (i, j) with a rotation angle θ , which is shown in Figures 4(b) and (c). A local image patch I , centred at the location with the rotation angle, is then extracted as the current environment of the tracker. The key problem of tracking is then formulated as a neural network $f(x)$ that learns a probability distribution as follows:

$$f(x) = p(\varphi|\Phi(I(i, j), \Delta z)), \quad (1)$$

where Δz is the dynamic window size and Φ is the function of preprocessing.

We further simplify the turning angle φ as four discrete cases in the reactive unit of FSM (shown in Figure 8) and implement a DNN to learn this probability distribution. This formulation significantly reduces the complexity of the learning problem (compared with the problem of recognition of road and other objects).

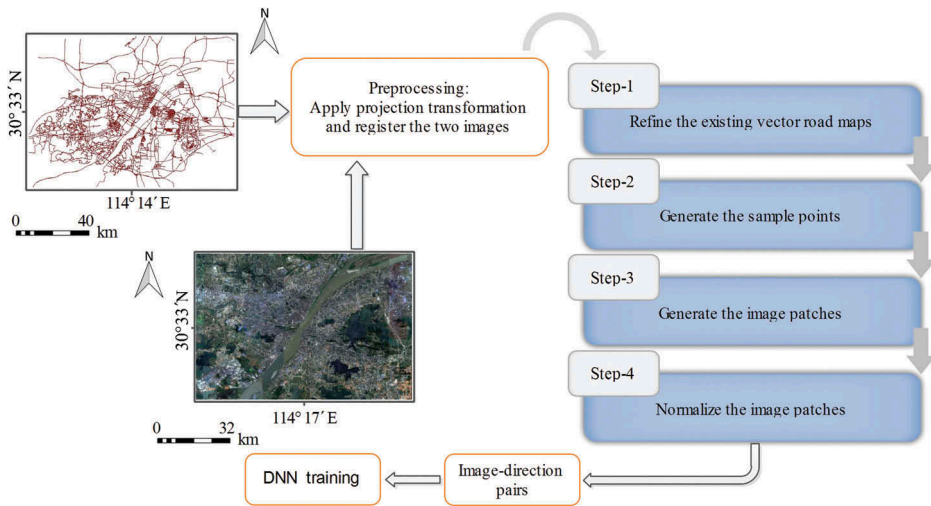


Figure 3. Image-direction pairs are extracted by combining vector road maps and imagery. These labelled image patches will be used to train our DNN.

2.2. Training data set preparation

Different from the generation of road and non-road data sets (Mnih and Hinton 2010), this study follows the purpose illustrated in the problem formulation section and proposes a vector-guided sampling strategy. Here, image-direction pairs are extracted by combining the imagery and the vector road map. A flow chart is shown in Figure 3 and the strategy is outlined as follows:

2.2.1. Preprocessing

Vector road maps can be obtained from various sources, e.g. transport agencies, mapping departments, or online map websites. Many different spatial references are commonly used by vector road maps and the imagery. Before the imagery is correctly and precisely overlaid with the vector road maps, projection transformation and image registering are applied to the two data sets.

2.2.2. Vector refine

The existing vector road maps were usually produced by operators, and the production is very time consuming: a vertex means a click of the mouse of the operator. For the sake of efficiency, there are insufficient vertices where road turns sharply, such as points A and B in Figure 4(a). To make the morphological characteristics of roads more visually matched between the polyline and the image, we employ cubic B-splines to refine these vectors. Subdivision directly to the curve is very difficult. Instead, the B-spline divides the domain of the curve. A set of non-decreasing numbers is defined as: $\mu_0 \leq \mu_1 \leq \mu_2 \dots < \mu_n$. These μ_k ($k \in (0, n)$) in set U can be considered to be division points or knots. All basis functions of B-splines should be defined on the interval of $[\mu_0, \mu_n]$ with a degree d . The Cox-de Boor recursion formula is a commonly used algorithm generating basis functions and defined as follows:

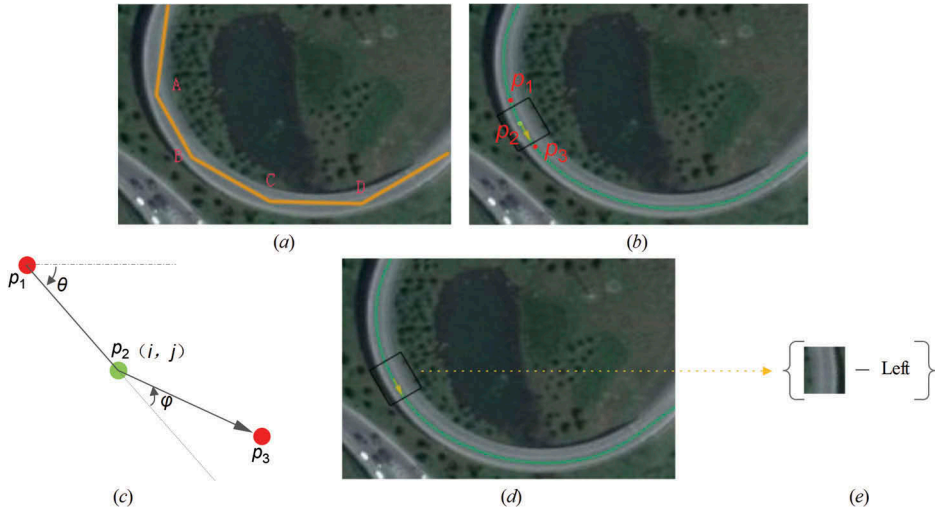


Figure 4. Vector-guided sampling strategy. (a) The original vector map (orange polyline) overlaid with the image. (b) Refined vector. (c) Geometrical relationship when a tracker locates in point p_2 . (d) An illustration of tracker with an orientation (yellow arrow) and detection window (black rectangle). (e) An extracted image patch and a label.

$$N_{k,0}(0) = \begin{cases} 1 & \text{if } \mu_k \leq \mu \leq \mu_{k+1}, \\ 0 & \text{otherwise} \end{cases}, \quad (2)$$

$$N_{k,d}(\mu) = \frac{\mu - \mu_k}{\mu_{k+d} - \mu_k} N_{k,d-1}(\mu) + \frac{\mu_{k+d+1} - \mu}{\mu_{k+d+1} - \mu_{k+1}} N_{k+1,d-1}(\mu), \quad (3)$$

where μ is the knot of the curve.

Basis functions $N(\mu)$ with a degree of 3 and control points (nodes in the polyline) are combined to compute the every point on the final curve. As shown in Figure 4(b), the road segments with sharp turning are processed using the above algorithm and we can get more precise pairs from them later.

2.2.3. Sample point generation

Sample points are the centres of the image patches to be extracted. According the different types of scenes, we divide these points into five sets: P_{straight} , P_{left} , P_{right} , $P_{\text{intersection}}$, and P_{block} . First, we divide every road polyline by equal spaced points, such as: p_1 , p_2 , and p_3 in Figures 4(b) and (c). The turning angle φ is computed as in Equations (4) and (5):

$$\theta_i = \tan^{-1}((y_i - y_j), (x_i - x_j)), \quad (4)$$

$$\varphi = \begin{cases} \theta_i - \theta_j & \text{if } \theta_i \theta_j \geq 0 \\ \min((\theta_i - \theta_j), (2\pi + \theta_j - \theta_i)) & \text{otherwise} \end{cases}, \quad (5)$$

where (x_i, y_i) and (x_j, y_j) are the coordinates of consecutive points p_i and p_j , e.g. p_2 and p_3 . If $|\varphi| < T_1$ (T_1 is a threshold and set to be 8° in experiments), we believe that p is located in

the straight road segment and should be added to set P_{straight} . If $\varphi > T_1$, p is located in the right turning road segment and should be added to set P_{right} . For the same reason, we add p to set P_{left} if $\varphi < -T_1$.

Let L_1 be a polyline, L_2 another polyline, if $p \in L_1 \cap L_2$, then add p to $P_{\text{intersection}}$. P_{block} are the locations where image patches are considered non-roads and to be blocked. Mnih and Hinton (2010) have provided a method to generate non-road samples. In Equation (6), $\psi(i, j)$ can be interpreted as probability that a point p belongs to road, where $d(i, j)$ is the Euclidean distance between the point p and the nearest centre line of roads. The smoothing parameter δ is set such that the width of $2\delta + 1$ pixels approximately equals to that of a typical two-lane road (Mnih 2013) and a threshold T_2 is defined to measure $\psi(i, j)$. We add p to set P_{block} when point p is consistent with $\psi(i, j) < T_2$.

$$\psi(i, j) = e^{-d^2(i, j)/\delta^2}. \quad (6)$$

2.2.4. Image patch generation

A local image patch is extracted based on the location $p_k \in \{P_{\text{straight}}, P_{\text{left}}, P_{\text{right}}, P_{\text{intersection}}, P_{\text{block}}\}$ and the rotation angle θ of the point. An affine transformation is then applied in a detecting window centred at p_k , the coordinate of which is denoted as (p_x, p_y) . The affine transformation matrix $\mathbf{A}$ is defined as

$$\mathbf{A} = \begin{pmatrix} \alpha\beta(1 - \alpha)p_x - \beta p_y \\ -\beta\alpha p_x + (1 - \alpha)p_y \end{pmatrix}, \quad (7)$$

where

$$\alpha = \lambda \cos \theta; \beta = \lambda \sin \theta, \quad (8)$$

with λ representing a scale factor.

An image patch and its corresponding direction tag are shown in Figure 4(d).

2.2.5. Normalization

The extracted image patches are highly susceptible to noisy, inconsistent data, as they come from different sensors or captured under different conditions. Before training, the data sets are preprocessed using methods proposed by Jarrett et al. (2009). The image patches in the labelled set are normalized locally in a 7×7 pixel window, ending up with unit standard deviation and zero mean. Previous research has shown that the preprocessing approach is efficient in improving the recognition rate significantly (Opitz et al. 2014; Henaff et al. 2011).

2.3. DNN

Artificial neural networks were proposed in the early 1980s, but have been limited under some harsh conditions such as: the initial weights should be close enough to a good solution; the training data set should be big enough; the computing resource should be rich. DNNs have gone ahead by leaps and bounds since Hinton presented his instructive work (Hinton and Salakhutdinov 2006).

DNNs benefit from their multi-layers and learning hierarchically, which makes them more powerful in feature extraction and abstraction (Zeiler and Fergus 2014). Typical architectures of DNNs are deep belief neural networks, deep Boltzmann machines, stacked autoencoder, and DNN. Instead of defining task-specific features such as shape index (SI) and compactness (COM) (Huang and Zhang 2009), edge map (Das, Mirmalinee, and Varghese 2011; Gang, Jinliang, and Chen 2011), feature extraction is included in the weight optimization in DNNs, which means that the arbitrary feature extraction method need not be specified by trial and error.

Recently, DNNs have achieved near-human performance in many recognition tasks such as handwriting, traffic signs, and face recognition (Ciresan, Meier, and Schmidhuber 2012; Krizhevsky, Sutskever, and Hinton 2012; Swietojanski, Ghoshal, and Renals 2014; Opitz et al. 2014). Many research studies employ DNNs in optical and hyperspectral remote-sensing images and show great potential in dealing with the classification problem in remote-sensing applications (Chen et al. 2014; Song et al. 2013).

This study focuses on the DNNs and trains one to recognize the direction of roads from input context, as shown in Figures 5 and 6. Many good features, such as max pooling and receptive fields, make the DNN deeper (with more layers) and wider (with more maps). A DNN has an input layer with $N \times N$ neurons, many convolutional layers, each of which followed by a max-pooling layer, and a fully connected layer. The max-pooling layers lie between convolution layers. A simple max pooling technique determines winning neurons by partitioning convolution layers into quadratic regions of local inhibition, selecting the most active neurons of each region. The winners of some layer represent a smaller, down-sampled layer with lower resolution, feeding the next layer in the hierarchy. Thus, the pooling is a way of sub-sampling, reducing the dimension of feature vectors in convolution layers. In Figure 5, take the first and second convolutional layers for example. The first convolutional layer has maps with 44×44 neurons. With the help of one max-pooling layer, the dimensions of the maps are reduced to 20×20 neurons.

For the deep architecture, the max-pooling layers have a more profound significance. The max-pooling layer reserves the most responsive (higher activation values) neurons by partitioning the convolutional layer into quadratic regions of local inhibition, which is motivated by the biological research that human brains are likely to use such a neural activation structure. The winners of the layers feed the next layer in hierarchy, making the feature input transition independent. The successes of DNN on well-known data sets, e.g. face recognition and hand digit recognition, indicate that this way of pooling is meaningful and makes the deep architecture extract more richer features.

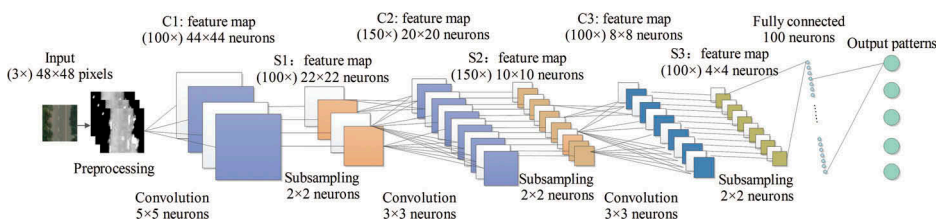


Figure 5. An architecture of DNN used in experiments.

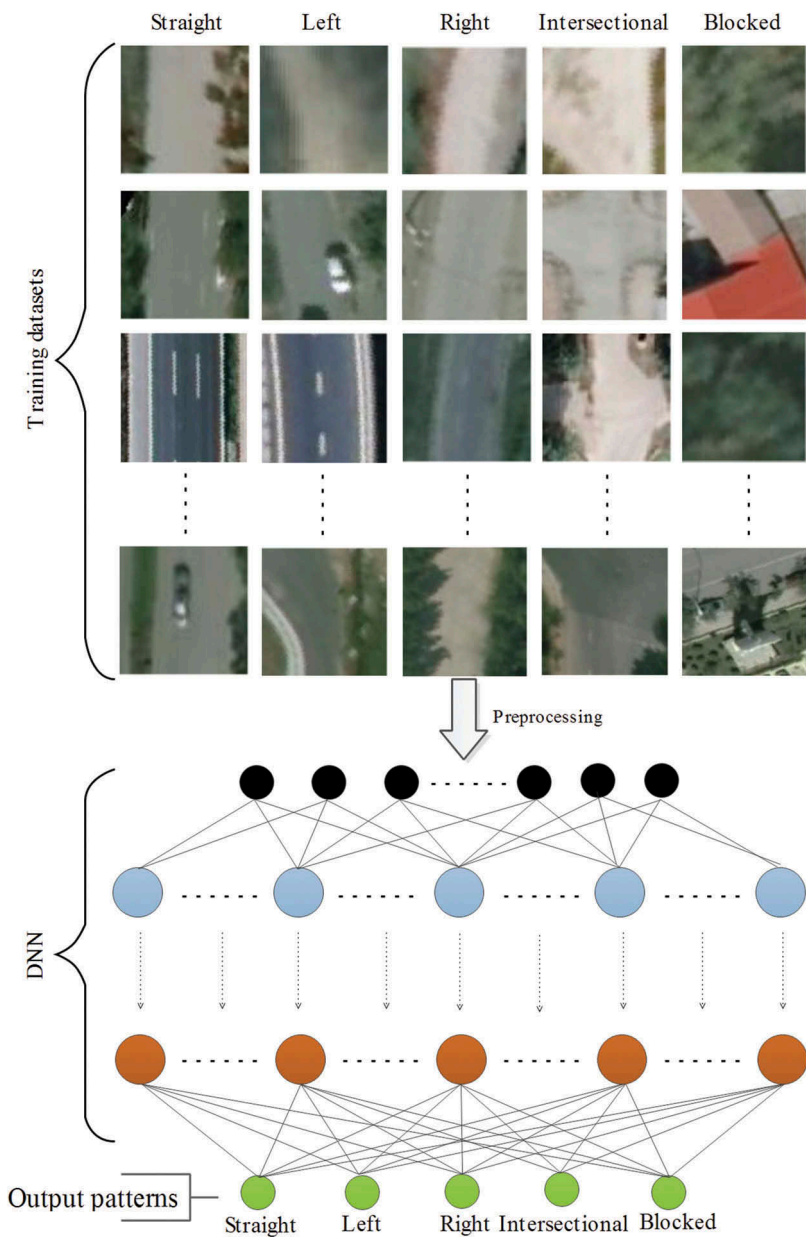


Figure 6. Training data sets include the five classes of image patches. DNN is trained to recognize patterns utilizing these abundant labelled pairs.

The DNN architecture used in experiments is shown in Figure 5. Before training, we preprocess the image patches by the approaches described above and map all pixels to the input layer.

In the forward propagation, the input of a neuron in the convolutional layer is a weighted summation in a rectangular section from previous layers, as defined in Equation (9). Every rectangular cell in the same layer shares a convolution filter ω , the initial value

of which is obtained from a uniform random distribution in the range $[-0.05, 0.05]$ (Ciresan, Meier, and Schmidhuber 2012). Generally, the activation is a non-linear function. Max-pooling layers reduce the dimension of the problem by introducing sparseness. In our networks, a $k \times k$ neuron block is reduced to single value. The output of fully connected layer is relating to the category, i.e. the five predefined direction types of roads.

$$x_{i,j}^l = \sum_{a=0}^m \sum_{b=0}^m \omega_{a,b} y_{(i+a),(j+b)}^{l-1}, \quad (9)$$

where $x_{i,j}^l$ is the input for the (i, j) neuron of the l th layer, $y_{i,j}^l$ is the activation value associated with $x_{i,j}^l$, and $\omega_{a,b}$ is the convolution filter of $m \times m$ dimensions for one layer.

In the backward propagation, an error function E is defined between the output and the ground truth. The goal of training is minimizing it by optimizing the weight matrix $\mathbf{W}$. The weight is updated with a rule:

$$\mathbf{W} = \mathbf{W} - \eta \frac{\partial E}{\partial \mathbf{W}}, \quad (10)$$

where η is the learning rate with an initial value η_0 and a decreasing rate μ .

All contribution of $\mathbf{W}$ is summed in computation of x and a chain rule is thus applied to compute the gradient components. For the same reason, we get the gradient components over x in the current layer and the gradient components over y in the previous layer. Repeat Equation (11) until all errors (gradient components in E) are known.

$$\frac{\partial E}{\partial \omega_{a,b}} = \sum_{i=0}^{N-m} \sum_{j=0}^{N-m} \frac{\partial E}{\partial x_{i,j}^l} \frac{\partial x_{i,j}^l}{\partial \omega_{a,b}} = \sum_{i=0}^{N-m} \sum_{j=0}^{N-m} \frac{\partial E}{\partial x_{i,j}^l} y_{(i+a),(j+b)}^{l-1}. \quad (11)$$

where N is number of neurons in one layer.

After training the networks, hidden layers with convolution and max-pooling can learn features hierarchically and extract deeper features from image patches of the environment.

2.4. Road tracking

2.4.1. FSM

FSM based approaches are popular in Robotics (Medeiros 1998), such as autonomous mobile systems (Kurt and Özgüner 2013). The mechanism of our tracker can be modelled by a FSM. The FSM is a mathematical model of computation that can be used to control a sequence of actions depending on triggering events. The machine should be in one of the finite states at a time. The FSM changes its state when a satisfactory event occurs, which is another property of FSM: state transition. An FSM can be defined mathematically as a quintuple (K, S, S_0, C, S_f) . K is a set of events, which act as the inputs of the FSM. S is the predefined, finite, and non-empty set of states. C is a state-transition function: $C = K \times S$. S_0 are the initial states and S_f are the final states, which are the subsets of state set S .

The states of the FSM are associated with output patterns of the DNN. We define the state set $S = \{\text{straight, left turn, right turn, intersection, block}\}$, and define the event set $K = \{\text{straight pattern, left pattern, right pattern, intersectional pattern, blocked pattern}\}$.

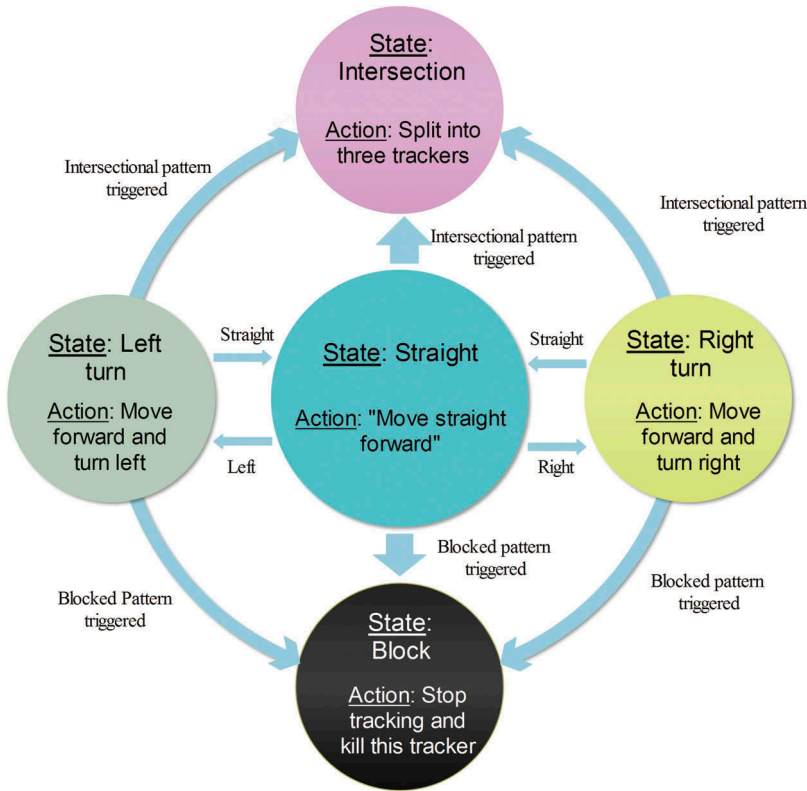


Figure 7. States, transitions, and actions in the FSM.

Figure 7 shows the states, transitions, and actions in the FSM. States and corresponding actions are shown in circles. The connections, denoted by coloured lines, are state transitions, which are triggered by the triggering events, shown near the connections. There are 5 states and 12 state transitions (including transitions between ‘left turn’ and ‘right turn’) in the FSM. The states: ‘block’ and ‘intersection’ are in the final states set S_f . As deterministic FSMs, the FSM will lead to only one state when given an input from a particular state. The evidence for the transitions depends on more than one image patches. When a new pattern is detected, n ($n > 2$) image patches will be extracted in consecutive locations and analysed in the DNN. Whether the state transition occurs is decided by the result of majority voting described in the next section.

The reactive unit in FSM is associated with the states and directly responsible for the motion control of the tracker. When receiving the current states s (states in the set S), the reactive unit implements a local behaviour for the tracker, e.g. ‘go straight’ or ‘turn right’. We define five reactions in the unit as follows:

- (1) If $s = \text{‘straight’}$, the tracker advances with a distance v , as shown in Figure 8(a). (2) If $s = \text{‘left turn’}$, the tracker moves straight forward with a distance v and then turns left with an angle α . As shown in Figure 8(b), a tank-style turning is applied in experiments.
- (3) If $s = \text{‘right turn’}$, the reaction share a similar way to case (2). (4) If $s = \text{‘intersection’}$, the current tracker is split into three trackers with three orientations, the current states of

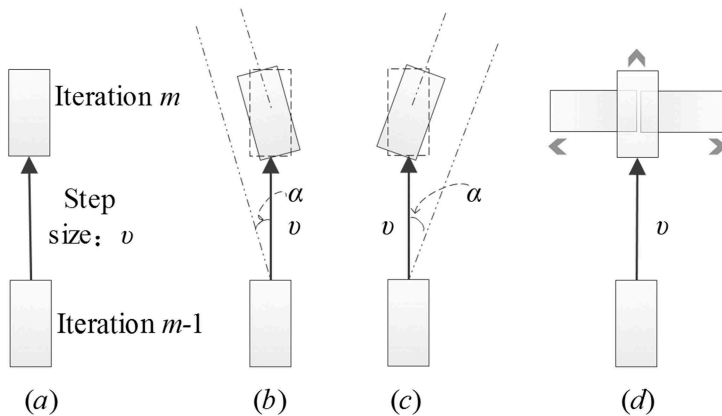


Figure 8. Four reactive cases: (a) straight; (b) left turn; (c) right turn; (d) intersection. The trackers are denoted by the rectangles, the relative positions of which in adjacent iterate steps manifest the effects of the reactive unit.

which are, respectively, set to 'straight', 'right turn', and 'left turn', as shown in Figure 8(d). (5) If $s = \text{'block'}$, the tracker stops tracking.

One thing needs to be specified is that step size v and turning angle α are small enough for the fact that all kinds of turns (sharp or soft) or intersections can be divided by small turning angles and steps. There are two principles in setting the values of v and α . The most important one is that the tracker is configured to trace in any curve to conform the various shapes of roads. The 90° turning is considered to be a hard case in road tracking and taken as an example here. As shown in Figure 9, the dashed lines denote the trajectories of the trackers, which start to turn right at point A with three different configurations. The trackers keep turning until they pull into the road in the horizontal direction. The three trackers are configured as: T_1 ($v = 2$ pixels, $\alpha = 5^\circ$), T_2 ($v = 2$ pixels, $\alpha = 8^\circ$), and T_3 ($v = 1$ pixel, $\alpha = 8^\circ$). Visually, the second tracker does a good job while the first and third fail. However, it should be noticed that this is an extreme case, where the trackers always keep turning right. In the real system, the tracker will often receive a 'straight pattern triggered' signal and the FSM will be in the 'straight' state, in which, the tracker will stop turning and go straight for steps. In other words, the tracker will adjust the orientation automatically according to the current context and should not fail in all the three cases (T_1 , T_2 , T_3). In experiments of road extraction, we test the tracker with the three configurations and all the combinations produce good results with subtle difference. A broad range (combinations) of values of v and α can be accepted if only we keep them small enough (e.g. $v < 3$ pixels, $\alpha < 10^\circ$).

The other principle is that the tracing should be as efficient as possible. In practice, we implement another option for testing: the real tank-style turning ($v = 0$ pixel, $\alpha = 8^\circ$), where the tracker will keep turning without moving forward. The tracker moves forward until the FSM is in the 'straight' state. Theoretically, this should be the best as it can conform to more complex curves. However, this configuration is time consuming and unnecessary. After going through several experiments, we chose ($v = 2$ pixels, $\alpha = 8^\circ$) as the default configuration in our system.

The tracing route after tracking can be described by combining the sequencing states in the FSM and locations of the tracker; an example is shown in Figure 12(c). In one

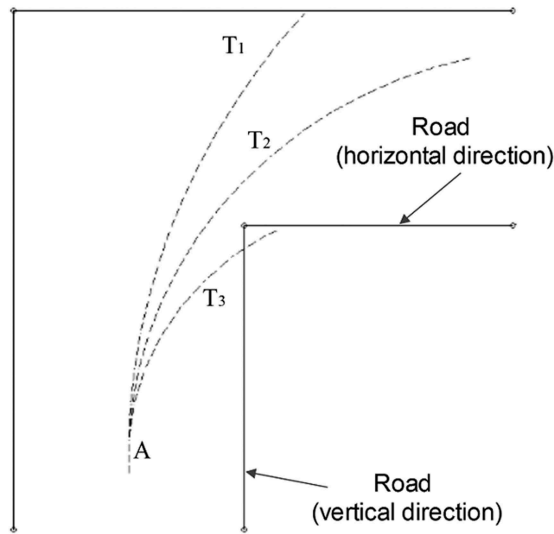


Figure 9. A 90° turning case with three different configurations.

word, the FSM makes our framework be able to deal with: inputs (image patches) that is the evidence for changing from one state to another, and actions, each of which is associated with specific states, making the tracker be capable of reacting properly to follow the roads.

2.4.2. Detection window size optimization

Due to the varying widths of the roads in remote-sensing imagery, it is important to select an optimal scale (sample window size) for road direction recognition. The geostatistical analysis proves that there is no single scale (fixed window size) that would completely describe the features of roads in VHR imagery (Coburn and Roberts 2004). The multiscale approach makes the recognition more robust for roads with different widths, which is utilized in this study.

Let s_0 be the current state and s_1 be the next candidate state. According to the state-transition rules, if pattern κ (detected pattern of the DNN) is consistent with the state s_0 , e.g. κ is a 'straight pattern' and s_0 is a 'straight' state, the FSM will directly accept the pattern κ and retain the current state. Otherwise, e.g. κ is a 'right pattern' and s_0 is a 'straight' state, the transition may occur. However, as it has been illustrated in the previous section, the transition will not depend on only one judgement from one location and one scale. The authors implement verification for this transition utilizing majority voting. Image patches with multiscales at consecutive locations are extracted, which will be analysed in DNN. Let function $O(m, n)$ be the output of DNN at the m th scale (total $1 - \Delta m$ scales (Δm denotes the number of scales)) and n th location (total $1 - \Delta n$ locations (Δn denotes the number of consecutive locations)). We define another function $G(m, n)$ as follows:

$$G(m, n) = \begin{cases} 1 & O(m, n) = \kappa \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

A majority voting, defined in Equation (13), is used to integrate the multiscale information at the decision level. If G_m is not smaller than $\Delta n/2$, the FSM accepts the pattern κ and the transition occurs. At the same time, the detecting scale of the next state will be changed to scale m . Three scales and four consecutive locations are used in the experiments.

$$G_m = \sum_{n=1}^{\Delta n} G(m, n). \quad (13)$$

3. Experimental results and discussion

In experiments, we first trained the DNN with an extensive amount of training data sets. Scenes from satellite and aerial images are subsequently used for evaluating the performance of our tracker.

3.1. Training the tracker

We generate 500,000 samples (image-direction pairs) from imagery and vector road maps in Wuhan, China. The area has the longitude ranging from $113^{\circ} 41'$ E to $115^{\circ} 5'$ E, and the latitude ranging from $29^{\circ} 58'$ N to $31^{\circ} 22'$ N. Two types of imagery in the area are used: aerial imagery and Google Earth imagery that shows great potential in land-use related research (Hu et al. 2013). The Google earth imagery, obtained on 17 December 2010, was originally taken from the QuickBird imagery. The aerial imagery was captured using a Leica ADS80 airborne digital sensor on 6 March 2012. The vector road maps are downloaded from the Open Street Map (Haklay and Weber 2008). Only three bands: red, green, and blue, are used in the experiment. There are about 100,000 samples for each pattern, which cover almost all kinds of roads, e.g. highways, country roads, and urban roads. Part of the training data sets is shown in Figure 10. These image patches are generated automatically by the proposed method. The samples of 'left', 'right', 'straight', 'intersectional', and 'blocked' are shown roughly in the top left, top right, centre, bottom left, and bottom right, respectively. Some of the samples are in bigger sizes intended to show the details in different clusters while the small ones represent the richness of the whole data set. These labelled data sets are randomly divided into 60% for training data sets, 20% for validation data sets, and 20% for test data sets.

The DNN consists of the following layers: an input layer with 3 maps, a convolutional layer with 100 maps and 5×5 neuron filters, a max-pooling layer with 2×2 neuron non-overlay windows, a convolutional layer with 150 maps and 3×3 neuron filters, a max-pooling layer with 2×2 neuron non-overlay windows, a convolutional layer with 100 maps and 3×3 neuron filters, a max-pooling layer with 2×2 neuron non-overlay windows, a fully connected layer with 100 hidden units, and a fully connected output layer with 5 neurons (patterns).

An annealed learning rate is used with an initial value of 0.001, a declining rate of 0.994 per epoch, and an end value of 0.00002, which is a useful strategy for reducing overfitting (Der, Martius, and Pfeifer 2012).

Due to the massive training data sets, the depth, and numerous maps of the DNN, we spend 10 h for only one epoch of the training data sets, when trying to implement the DNN on a 4-core central processing unit (CPU). We finally implement the DNN on a machine with a consumer graphical processing unit (GPU) (GeForce GTX 660) based on

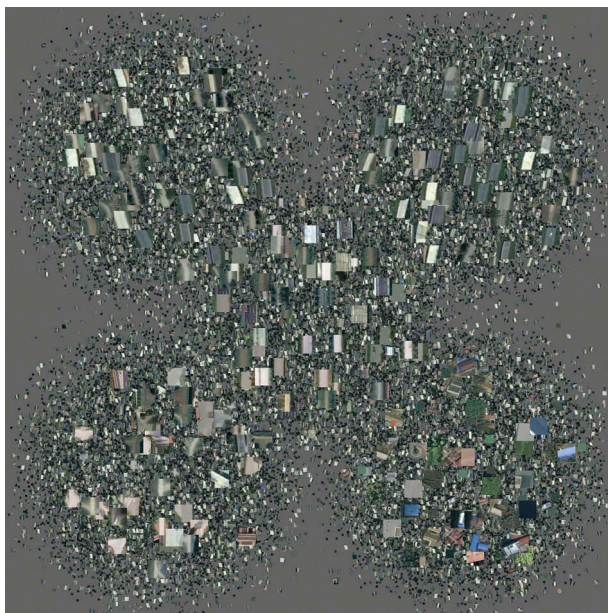


Figure 10. Part of the labelled data set in a clustered view. The samples of ‘left’, ‘right’, ‘straight’, ‘intersectional’, and ‘blocked’ are shown roughly in the top left, top right, centre, bottom left, and bottom right, respectively.

Theano, a Python library for deep learning (Bastien et al. 2012). The training procedure is significantly accelerated and finished after 900 epochs in 60 h.

The average errors on the separated test data sets achieve 1.323% and the confusion matrix is shown in Figure 11. The confusion matrix demonstrates that the trained DNN almost perfectly recognize the predefined five patterns. Most of the error occurs between straight and right/left because some turnings are too slight, although humans easily distinguish almost all the slight ones. Our DNN reaches a competitive performance and it is robust enough to analysis the environment of the tracker in the next section.

3.2. Extracting roads from imagery using the trained tracker

In this article, the performance of the proposed method is evaluated with challenging scenes from different regions. Figure 12 shows the process of road tracking of the proposed method. The aerial image, shown in Figure 12(a), was acquired using an Intergraph/ZI digital mapping camera (DMC) by the company Rhine-Westphalia Electricity (RWE) Power on 24 July and 6 August 2008. This test site locates in the Vaihingen, Germany, with a left top corner of $48^{\circ} 56' 03''$ N and $8^{\circ} 57' 20''$ E. It is a subset of data set 1 described below. The initial point of the tracker is also shown in Figure 12(b). Figure 12(b) shows the tracing route with different colours denoting states of the FSM. A tree structure, as shown in Figure 12(c), is used to present the whole states and transitions in the FSM. In both Figures 12(b) and (c), cyan denotes the state ‘straight’, blue the state ‘left turn’, yellow the state ‘right turn’, pink the state ‘intersection’, and black the state ‘blocked’ (these states are illustrated in Section 2.4.1). The results

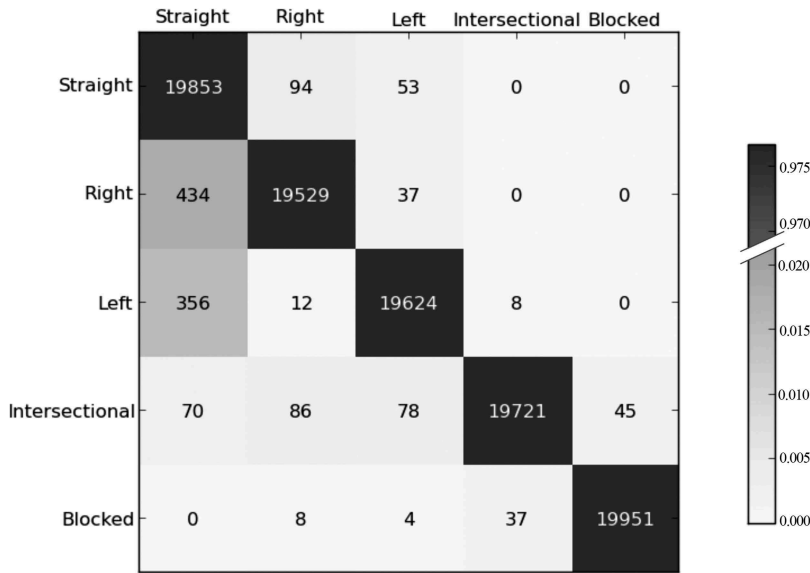


Figure 11. Confusion matrix of the test data set.

demonstrate that the DNN-embedded tracker can accurately estimate the direction of the roads. Figure 12(d) shows the final extracted road structure.

Based on the work of Wiedemann and Ebner (2000), completeness, correctness, and quality index are used to evaluate the performance of our results. Completeness is related to omission errors and defined as follows:

$$P_{\text{completeness}} = \frac{L_T}{L_T + L_N}, \quad (14)$$

where L_T is the length extracted using the tracker and L_N is the length of the omitted road. $P_{\text{correctness}}$, defined in Equation (15), is related to commission errors.

$$P_{\text{correctness}} = \frac{L_T}{L_T + L_E}, \quad (15)$$

$$P_{\text{quality}} = \frac{L_T}{L_T + L_N + L_E}, \quad (16)$$

where L_E is the length of the extracted roads that are not actually roads. In Equation (16), P_{quality} is the ratio of true positives over the sum of true positives, false positives, and false negatives.

As shown in Figure 13, four different categories of images, covering the urban, suburban, and agricultural areas, are used to test the robustness and flexibility of the proposed algorithm. Figures 13(a) and (b) show another two chips of images in data sets 1 and 2, respectively. Data sets 1 and 2 are aerial images of less than 0.5 m/pixel and provided by International Society for Photogrammetry and Remote Sensing (ISPRS) working group III/4, which aims at making progress in the automatic recognition of

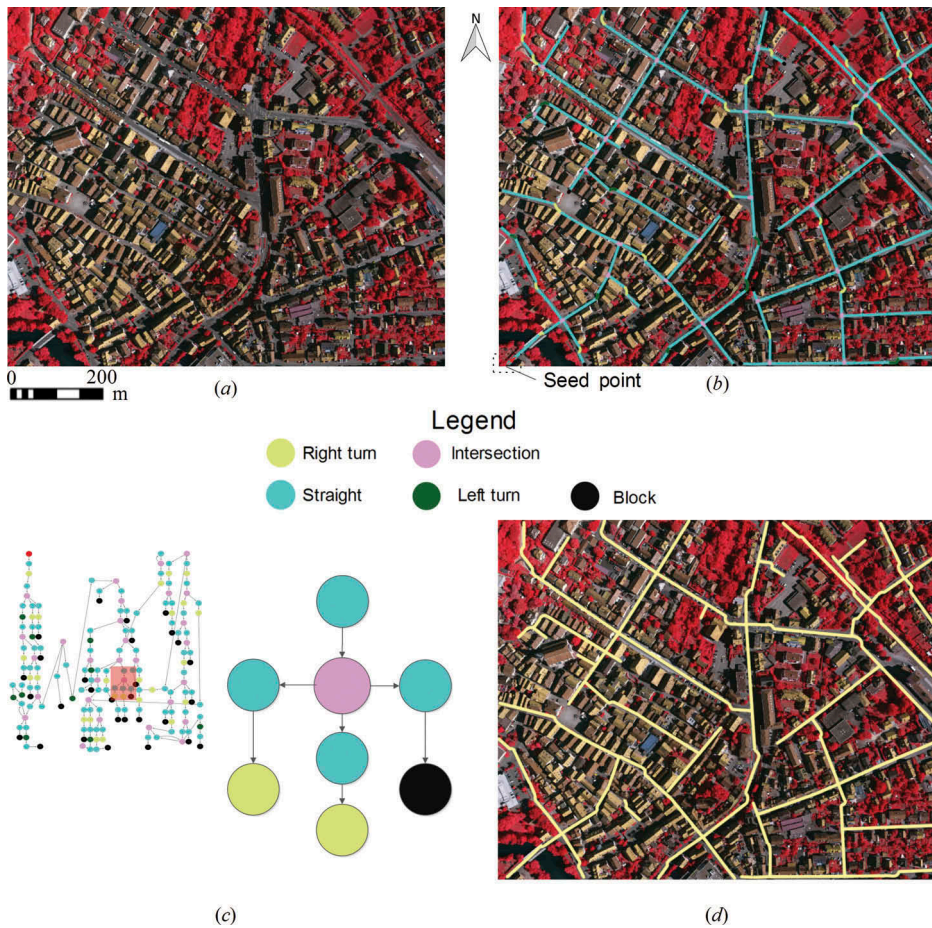


Figure 12. Road tracking using the proposed method. (a) Original aerial image of 0.2 m/pixel. (b) Extracted road networks with different colours denoting corresponding states in the FSM. (c) A tree structure showing the states and transitions of FSM during the above extraction. (d) Extracted road networks denoted by yellow lines overlaid on the original image.

objects in complex scenes from images. Data sets 3 and 4 are images of 1 m/pixel and cropped from Wikimapia (Koriakine and Saveliev 2008). Figure 13(a) is characterized as residential area consisting of buildings and trees. A considerable part of the roads, shown in Figure 13(b), is shadowed by buildings, moreover, the roads and buildings share easily confusing spectral radiances. Figure 13(e) shows a scene, from data set 3, with flyovers which are usually arduous to extract in many proposed extraction methods. An agricultural scene, as shown in Figure 13(f) coming from data set 4, has roads with varying widths. The results in Table 1 demonstrate that the proposed method performs well on all the scenes.

As for comparison, we also implement this approach by adopting both state-of-the-art method from Poullis (2014) and a commercial tool Environment for Visualizing Images Feature Extraction (ENVI FX). Figure 14 shows a satellite image at 0.6 m spatial resolution. The scene covered by QuickBird in March 2014 for Las Vegas, USA, is characterized by its clear road networks and taken as one study area. Three bands (red,

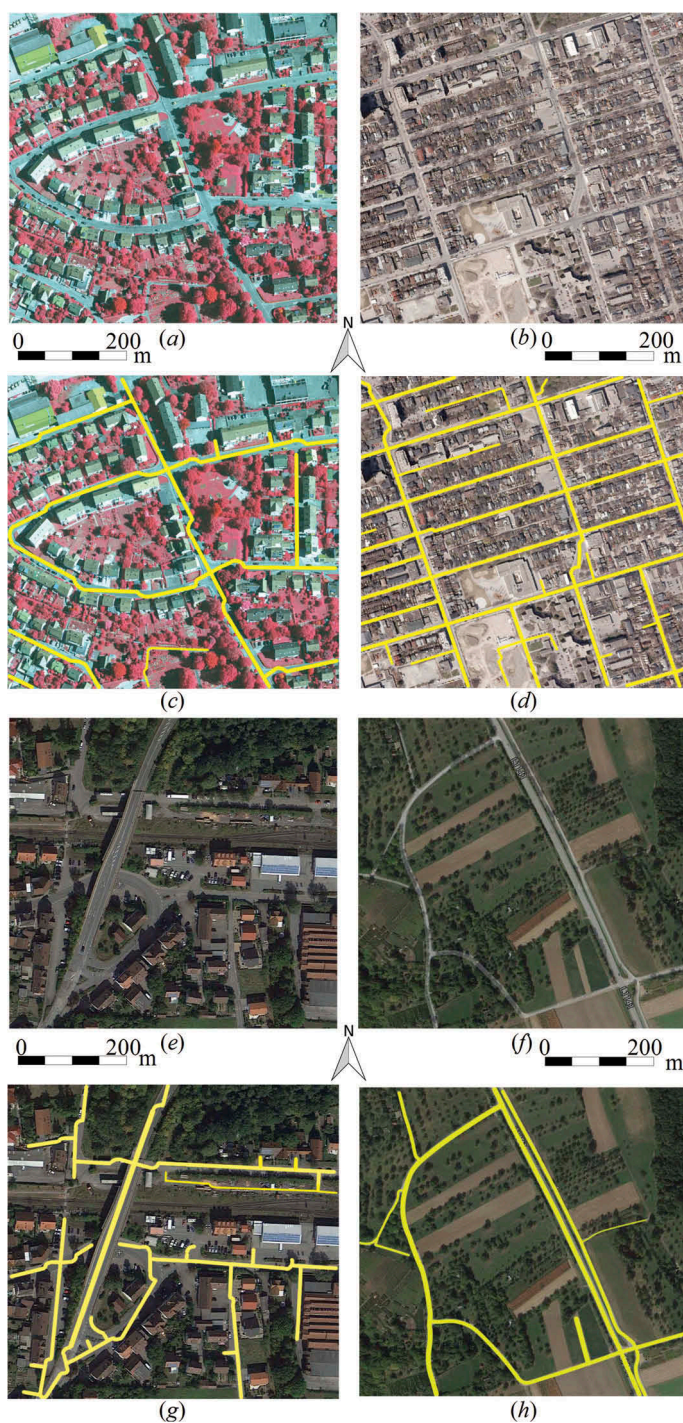


Figure 13. Experiments on four typical scenes. (a) An urban residential area. (b) A challenging scene where roads and bare lands share easily confusing spectral radiances. (e) A scene with flyovers, which is arduous to extract in previous methods. (f) An agricultural area with farmlands on both sides of roads. (c), (d), (g), and (h) are the corresponding results, with yellow lines denoting the extracted road networks.

Table 1. Quantitative evaluation for the four test data sets shown in Figure 13.

Data set	Resolution (m)	Completeness (%)	Correctness (%)	Quality (%)
Data set 1	0.1	74.4	83.2	72.3
Data set 2	0.2	80.8	81.4	79.8
Data set 3	1.0	87.2	93.3	92.7
Data set 4	1.0	91.4	93.1	91.9

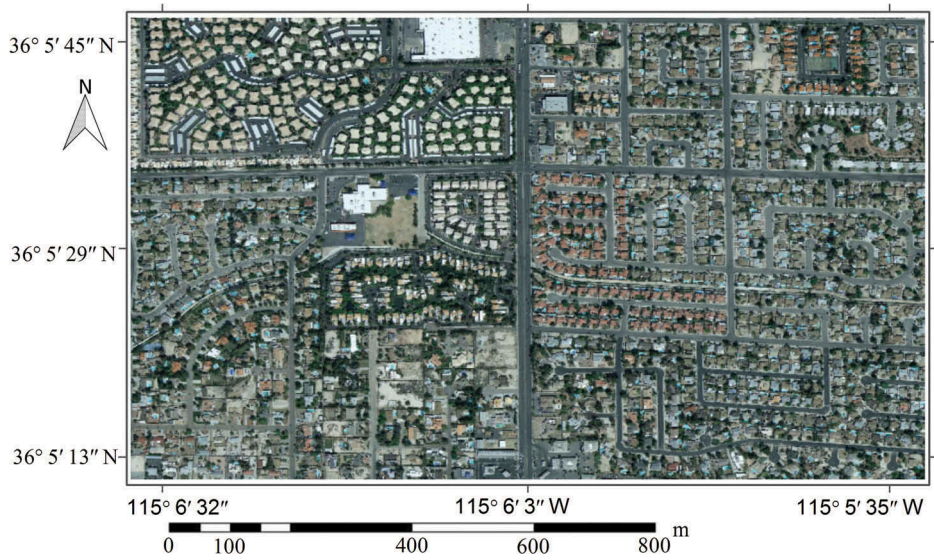


Figure 14. QuickBird image (R, G, B bands) with 0.6 m pixel size.

green, and blue bands) are utilized in the experiment. Figure 15 shows the result of road extraction from the image by applying the object-based and the proposed approaches. For visual and quantitative comparison of the methods, the manually interpreted road map, shown in Figure 15(d), is considered as ground truth. The evaluation of the performance on the study area is reported in Table 2 (test case 1).

Three test cases in different categories of lands are chosen from Das, Mirmalinee, and Varghese (2011). As shown in Figure 16, the satellite images at 1 m spatial resolution were acquired from Wikimapia. Performance of the road network extraction from the test cases is shown in Table 2 (test cases 2–4). It is observed that the proposed method performs very well in the three test cases. In the first image (column 1 in Figure 5), roads are not connected as it is captured in a small region, consequently two initial points are provided for the extraction in the proposed method. This will be rare in bigger images as Figure 15 or real applications of road extraction. Note that Das reports a slightly higher accuracy on these test cases. However, at least 14 data-driven parameters were tuned in the process. This study takes advantage of dynamic window detection and multiscale information fusion to eliminate the tuning of data-dependent thresholds in the process of road extraction.

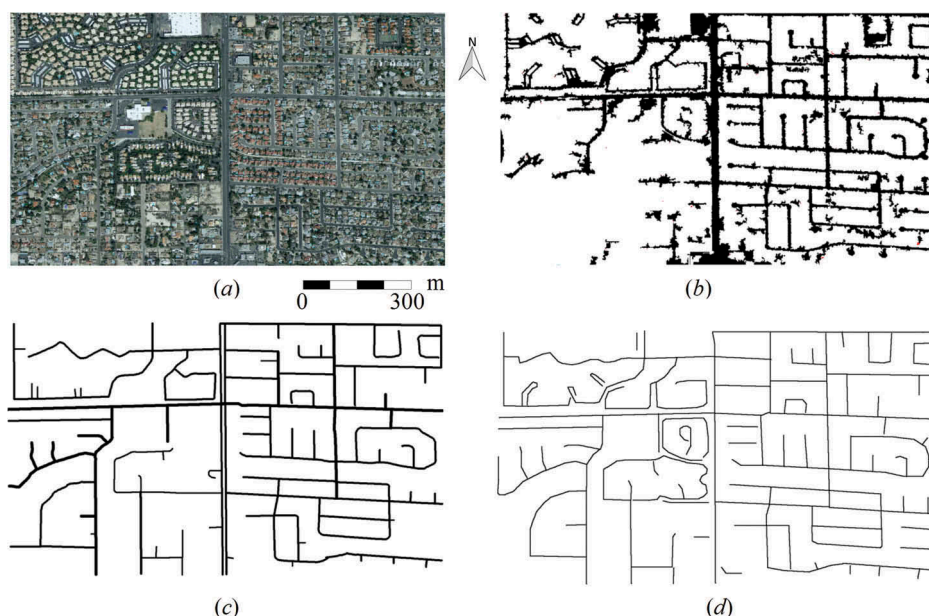


Figure 15. Road extraction of QuickBird data using the object-based and the proposed methods. (a) Input image. (b) Extraction results using the object-based method. (c) Extraction results using the proposed method. (d) A hand-drawn road map.

Figure 17(a) shows an input image of 0.6 m/pixel resolution acquired from Google Earth, the top left corner of the image is at $36^{\circ} 03' 52''$ N and $115^{\circ} 05' 31''$ E. As shown in Figure 17(b), the method proposed by Poullis (2014) excels on most of the scene but fails on some occasions marked by the rectangles. Parking lots and roads share the same intensity of radiation, and are arduous to separate, as shown in the lower-left rectangle. Inference of trees brings an interruption of road extraction in the upper rectangle. Many regular-located buildings are extracted as roads in the lower-right rectangle. Figure 17(c) shows the result of ENVI FX, which applies the object-based image analysis to extract roads. Apparently, the main structures are correctly extracted by the method. However, we must carefully adjust many parameters before extraction: one parameter for optimized linear, two for segment settings, two for merge settings, and eight for four chosen attributes (spectral mean, major length, texture various, and area). Furthermore, the adjustment uses an interactive way for choosing a better parameter, which is time consuming and exhausting. On the other hand, these parameters are data-variance, consequently, we have to rebuild our rules when processing images from other locations. Figure 17(d) shows the extracted road structure by the proposed method.

The quantitative evaluations for the results are shown in Tables 1–3. As it can be seen from Table 1, the proposed method performs quite well in these challenging areas and achieves acceptable results in VHR aerial imagery. Tables 2 and 3 show the results of comparative experiments; our method shows advantages over accuracy, no parameter turning. As the detection is completed within local context, the computation complexity increases linearly with the size of images. Accordingly, the proposed method also excels in processing time, which is marked in bold in Table 3.



Figure 16. Images from the data set in Das, Mirmalinec, and Varghese (2011). Row (a) represents the input images. Rows (b)–(d) show the extracted road networks by using the objected-based, Poullis (2014), and proposed methods, correspondingly.

For the proposed method, although the training process takes a lot of time, when completed, DNN judgement is timely to determine the direction of the image patches, approximately 3 ms for one decision. The entire tracking process needs to be triggered manually by selecting an initial position and orientation for the tracker. For remote-sensing images, each pixel is associated with specific geographic coordinates; therefore

Table 2. Performance of the road network extraction from the four test cases shown in Figures 15 and 16.

Image	Completeness (%)		Correctness (%)		Quality (%)	
	Object-based	Proposed	Object-based	Proposed	Object-based	Proposed
Test case 1	60	78	71	76	55	70
Test case 2	95	93	80	96	83	98
Test case 3	93	92	85	95	90	97
Test case 4	70	96	84	96	72	97



Figure 17. Comparison using the image data from Poullis (2014). (a) Original image of 0.6 m/pixel. (b) Results of Poullis. Interruptions of their methods caused by the inference of trees, buildings and parking lots are marked with red rectangles. Yellow lines denote the extracted road networks. (c) Results of ENVI FX. The extracted road surfaces are highlighted in green. (d) Results of proposed method. Extracted road networks are denoted by yellow lines.

Table 3. Comparison and quantitative evaluation for complex urban areas shown in Figure 17.

Method	Completeness (%)	Correctness (%)	Quality (%)	No. of parameters	Processing time (s)	Interaction required
Object-based	60	64	50	13	> 300	Parameter turning
Poullis (2014)	68	64	62	2	≈ 86	None
Proposed	75	70	74	None	<30	Seed point

the road networks extracted in one image can be used as seed points in the next matched image. Therefore, this semi-automatic method obtains a relative high degree of automation in applications.

The proposed approach performs well in non-very complex cases, where roads are clearly distributed on the images. The method seems not to perform well on too complex cases in aerial images, where roads and buildings share almost the same reflectance properties. Even though multiscales and consecutive locations are utilized to combat this, the output patterns of confusing image patches will be inaccurate and lead to wrong states in FSM.

4. Conclusions

Compared to the traditional learning-based methods, the proposed learning to track is a novel, useful and efficient way to extract road networks from the VHR imagery. Unlike previous studies that have focused on utilizing road-oriented low-level features to achieve a higher recognition rate, the authors have proposed a tracker exploiting the DNN to learn deep features directly from the image data. The proposed sample generation algorithm creates rich data set by combining imagery and existing vector road map. A large neural network with a deep architecture has been well trained to help recognize the directions of roads. The authors elaborately modelled the road tracker as a FSM, which associates its states with the output patterns of the DNN as a whole.

The results highlight three major conclusions. (1) The authors proposed a novel road tracker based on DNN. The tracker learns to track roads by training the DNN on large-scale sample data sets generated by a vector-guided sampling algorithm. (2) Road tracking process was modelled as a FSM that controls the behaviour of the tracker with recursive and reactive units, and encodes the path as a sequence of states and transitions. (3) The proposed fusion method makes the tracker detect with an optimized window size, which can obtain the primary road structure features better according to different road widths. The results presented herein indicate that the proposed approach could efficiently extract road networks from aerial and satellite VHR imagery.

In the future research, the authors plan to extend this method to automatically search for the initial candidate road points.

Acknowledgements

The authors would like to acknowledge the provision of the Downtown Toronto data set by Optech Inc., First Base Solutions Inc., GeoICT Lab at York University, and ISPRS WG III/4. The Vaihingen data set was provided by the German Society for Photogrammetry, Remote Sensing and Geoinformation (DGPF) (Jacobsen et al. 2010).

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by the High Resolution Earth Observation System of Major National Science and Technology projects [grant numbers 06-Y30B04-9002-13/15] and the National High Technology Research and Development Program of China [grant numbers 2012AA12A401, 2013AA12A403].

References

- Barsi, A., and C. Heipke. 2003. "Artificial Neural Networks for the Detection of Road Junctions in Aerial Images." *International Archives of Photogrammetry Remote Sensing and Spatial Information Sciences* 34 (3/W8): 113–118.
- Bastien, F., P. Lamblin, R. Pascanu, J. Bergstra, I. Goodfellow, A. Bergeron, N. Bouchard, D. Warde-Farley, and Y. Bengio. 2012. "Theano: New Features and Speed Improvements." NIPS 2012 Deep Learning Workshop, Lake Tahoe, November 23.
- Chai, D., W. Forstner, and F. Lafarge. 2013. "Recovering Line-Networks in Images by Junction-Point Processes." 2013 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Portland, OR, June 23–28, 1894–1901. doi:[10.1109/CVPR.2013.247](https://doi.org/10.1109/CVPR.2013.247).
- Chen, Y., Z. Lin, X. Zhao, G. Wang, and Y. Gu. 2014. "Deep Learning-Based Classification of Hyperspectral Data." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 7 (6): 2094–2107. doi:[10.1109/JSTARS.2014.2329330](https://doi.org/10.1109/JSTARS.2014.2329330).
- Cheng, J., W. Ding, X. Ku, and J. Sun. 2012. "Road Extraction from High-Resolution SAR Images via Automatic Local Detecting and Human-Guided Global Tracking." *International Journal of Antennas and Propagation* 2012: 1–10. doi:[10.1155/2012/989823](https://doi.org/10.1155/2012/989823).
- Ciresan, D., U. Meier, and J. Schmidhuber. 2012. "Multi-Column Deep Neural Networks for Image Classification." 2012 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Providence, RI, June 16–21, 3642–3649. doi:[10.1109/CVPR.2012.6248110](https://doi.org/10.1109/CVPR.2012.6248110).
- Coburn, C. A., and A. C. B. Roberts. 2004. "A Multiscale Texture Analysis Procedure for Improved Forest Stand Classification." *International Journal of Remote Sensing* 25 (20): 4287–4308. doi:[10.1080/0143116042000192367](https://doi.org/10.1080/0143116042000192367).
- Da Silva, C. R., and J. A. S. Centeno. 2012. "Semiautomatic Extraction of Main Road Centrelines in Aerial Images Acquired over Rural Areas." *International Journal of Remote Sensing* 33 (2): 502–516. doi:[10.1080/01431161.2010.540589](https://doi.org/10.1080/01431161.2010.540589).
- Das, S., T. T. Minalinee, and K. Varghese. 2011. "Use of Salient Features for the Design of a Multistage Framework to Extract Roads from High-Resolution Multispectral Satellite Images." *IEEE Transactions on Geoscience and Remote Sensing* 49 (10): 3906–3931. doi:[10.1109/TGRS.2011.2136381](https://doi.org/10.1109/TGRS.2011.2136381).
- Der, R., G. Martius, and R. Pfeifer. 2012. *The Playful Machine: Theoretical Foundation and Practical Realization of Self-Organizing Robots*. Berlin: Springer Science & Business Media.
- Gang, L., A. Jinliang, and C. Chen. 2011. "Automatic Road Extraction from High-Resolution Remote Sensing Image Based on Bat Model and Mutual Information Matching." *Journal of Computers* 6 (11): 2417–2426.
- Gerl, T. M., H. Kreibich, and M. Bochow. 2003. "Urban Structure Mapping using High-Resolution Remote Sensing Data for Modelling Flood Losses in Dresden, Germany." *EGU General Assembly Conference Abstracts* 15: 679.
- Haklay, M., and P. Weber. 2008. "Openstreetmap: User-Generated Street Maps." *IEEE Pervasive Computing* 7 (4): 12–18. doi:[10.1109/MPRV.2008.80](https://doi.org/10.1109/MPRV.2008.80).
- He, C., Z.-X. Liao, F. Yang, X.-P. Deng, and M.-S. Liao. 2012. "Road Extraction from SAR Imagery Based on Multiscale Geometric Analysis of Detector Responses." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 5 (5): 1373–1382. doi:[10.1109/JSTARS.2012.2219614](https://doi.org/10.1109/JSTARS.2012.2219614).
- Henaff, M., K. Jarrett, K. Kavukcuoglu, and L. Yann. 2011. "Unsupervised Learning of Sparse Features for Scalable Audio Classification." *International Society for Music Information Retrieval Conference* 11: 681–686.
- Hinton, G. E., and R. R. Salakhutdinov. 2006. "Reducing the Dimensionality of Data with Neural Networks." *Science* 313 (5786): 504–507. doi:[10.1126/science.1127647](https://doi.org/10.1126/science.1127647).
- Hu, J., A. Razdan, J. C. Femiani, M. Cui, and P. Wonka. 2007. "Road Network Extraction and Intersection Detection from Aerial Images by Tracking Road Footprints." *IEEE Transactions on Geoscience and Remote Sensing* 45 (12): 4144–4157. doi:[10.1109/TGRS.2007.906107](https://doi.org/10.1109/TGRS.2007.906107).
- Hu, Q., W. Wu, T. Xia, Q. Yu, P. Yang, Z. Li, and Q. Song. 2013. "Exploring the Use of Google Earth Imagery and Object-Based Methods in Land Use/Cover Mapping." *Remote Sensing* 5 (11): 6026–6042. doi:[10.3390/rs5116026](https://doi.org/10.3390/rs5116026).
- Huang, X., Q. Lu, and L. Zhang. 2014. "A Multi-Index Learning Approach for Classification of High-Resolution Remotely Sensed Images over Urban Areas." *ISPRS Journal of Photogrammetry and Remote Sensing* 90: 36–48. doi:[10.1016/j.isprsjprs.2014.01.008](https://doi.org/10.1016/j.isprsjprs.2014.01.008).

- Huang, X., and L. Zhang. 2009. "Road Centreline Extraction from High-Resolution Imagery Based on Multiscale Structural Features and Support Vector Machines." *International Journal of Remote Sensing* 30 (8): 1977–1987. doi:[10.1080/01431160802546837](https://doi.org/10.1080/01431160802546837).
- Huang, X., and L. Zhang. 2013. "An SVM Ensemble Approach Combining Spectral, Structural, and Semantic Features for the Classification of High-Resolution Remotely Sensed Imagery." *IEEE Transactions on Geoscience and Remote Sensing* 51 (1): 257–272. doi:[10.1109/TGRS.2012.2202912](https://doi.org/10.1109/TGRS.2012.2202912).
- Jacobsen, K., M. Cramer, R. Ladstädter, C. Ressler, and V. Spreckels. 2010. "DGPF-Project: Evaluation of Digital Photogrammetric Camera Systems—Geometric Performance." *Photogrammetrie-Fernerkundung-Geoinformation* 2010 (2): 83–97. doi:[10.1127/1432-8364/2010/0042](https://doi.org/10.1127/1432-8364/2010/0042).
- Jarrett, K., K. Kavukcuoglu, M. Ranzato, and Y. LeCun. 2009. "What is the Best Multi-Stage Architecture for Object Recognition?" IEEE 12th International Conference on Computer Vision, Kyoto, September 29–October 2, 2146–2153. doi:[10.1109/ICCV.2009.5459469](https://doi.org/10.1109/ICCV.2009.5459469).
- Koriakine, A., and E. Saveliev. 2008. "WikiMapia." Online: wikimapia.org.
- Krizhevsky, A., I. Sutskever, and G. E. Hinton. 2012. "Imagenet Classification with Deep Convolutional Neural Networks." Advances in Neural Information Processing Systems, Sierra Nevada, Lake Tahoe, November 23.
- Kurt, A., and Ü. Özgüner. 2013. "Hierarchical Finite State Machines for Autonomous Mobile Systems." *Control Engineering Practice* 21 (2): 184–194. doi:[10.1016/j.conengprac.2012.09.020](https://doi.org/10.1016/j.conengprac.2012.09.020).
- Li, X., S. Zhang, X. Pan, P. Dale, and R. Cropp. 2010. "Straight Road Edge Detection from High-Resolution Remote Sensing Images Based on the Ridgelet Transform with the Revised Parallel-Beam Radon Transform." *International Journal of Remote Sensing* 31 (19): 5041–5059. doi:[10.1080/01431160903283835](https://doi.org/10.1080/01431160903283835).
- Lin, X., J. Zhang, Z. Liu, J. Shen, and M. Duan. 2011. "Semi-Automatic Extraction of Road Networks by Least Squares Interlaced Template Matching in Urban Areas." *International Journal of Remote Sensing* 32 (17): 4943–4959. doi:[10.1080/01431161.2010.493565](https://doi.org/10.1080/01431161.2010.493565).
- Liu, J., H. Sui, M. Tao, K. Sun, and X. Mei. 2013. "Road Extraction from SAR Imagery Based on an Improved Particle Filtering and Snake Model." *International Journal of Remote Sensing* 34 (22): 8199–8214. doi:[10.1080/01431161.2013.835082](https://doi.org/10.1080/01431161.2013.835082).
- McKeown, J., M. David, and J. L. Denlinger. 1988. "Cooperative Methods for Road Tracking in Aerial Imagery." Proceedings CVPR' 88, Computer Society Conference on Computer Vision and Pattern Recognition, Ann Arbor, MI, June 5–9, 662–672. doi:[10.1109/CVPR.1988.196307](https://doi.org/10.1109/CVPR.1988.196307).
- Medeiros, A. A. D. 1998. "A Survey of Control Architectures for Autonomous Mobile Robots." *Journal of the Brazilian Computer Society* 4: 3. doi:[10.1590/S0104-65001998000100004](https://doi.org/10.1590/S0104-65001998000100004).
- Miao, Z., B. Wang, W. Shi, and H. Wu. 2014. "A Method for Accurate Road Centerline Extraction from a Classified Image." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 7 (12): 4762–4771. doi:[10.1109/JSTARS.2014.2309613](https://doi.org/10.1109/JSTARS.2014.2309613).
- Mnih, V. 2013. *Machine Learning for Aerial Image Labeling*. Toronto, ON: University of Toronto.
- Mnih, V., and G. E. Hinton. 2010. "Learning to Detect Roads in High-Resolution Aerial Images." In *Computer Vision ECCV 2010*, Vol. 6316, 210–223. Berlin: Springer-Verlag. doi:[10.1007/978-3-642-15567-3](https://doi.org/10.1007/978-3-642-15567-3).
- Montoya-Zegarra, J. A., J. D. Wegner, E. Ladický, and K. Schindler. 2014. "Mind the Gap: Modeling Local and Global Context in (Road) Networks." *Pattern Recognition* 8753: 212–223. doi:[10.1007/978-3-319-11752-2_17](https://doi.org/10.1007/978-3-319-11752-2_17).
- Opitz, M., M. Diem, S. Fiel, F. Kleber, and R. Sablatnig. 2014. "End-to-End Text Recognition Using Local Ternary Patterns, MSER and Deep Convolutional Nets." 11th IAPR International Workshop on Document Analysis Systems (DAS), Tours, April 7–10, 186–190. doi:[10.1109/DAS.2014.29](https://doi.org/10.1109/DAS.2014.29).
- Peng, T., I. H. Jermyn, V. Prinet, and J. Zerubia. 2008. "Incorporating Generic and Specific Prior Knowledge in a Multiscale Phase Field Model for Road Extraction from VHR Images." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 1 (2): 139–146. doi:[10.1109/JSTARS.2008.922318](https://doi.org/10.1109/JSTARS.2008.922318).
- Poullis, C. 2014. "Tensor-Cuts: A Simultaneous Multi-Type Feature Extractor and Classifier and Its Application to Road Extraction from Satellite Images." *ISPRS Journal of Photogrammetry and Remote Sensing* 95: 93–108. doi:[10.1016/j.isprsjprs.2014.06.006](https://doi.org/10.1016/j.isprsjprs.2014.06.006).

- Shao, Y., B. Guo, X. Hu, and L. Di. 2011. "Application of a Fast Linear Feature Detector to Road Extraction from Remotely Sensed Imagery." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 4 (3): 626–631. doi:[10.1109/JSTARS.2010.2094181](https://doi.org/10.1109/JSTARS.2010.2094181).
- Singh, P. P., and R. D. Garg. 2014. "A Two-Stage Framework for Road Extraction from High-Resolution Satellite Images by Using Prominent Features of Impervious Surfaces." *International Journal of Remote Sensing* 35 (24): 8074–8107. doi:[10.1080/01431161.2014.978956](https://doi.org/10.1080/01431161.2014.978956).
- Song, H., R. Xu, Y. Ma, and G. Li. 2013. "Classification of ETM+ Remote Sensing Image Based on Hybrid Algorithm of Genetic Algorithm and Back Propagation Neural Network." *Mathematical Problems in Engineering* 2013: 1–8. doi:[10.1155/2013/719756](https://doi.org/10.1155/2013/719756).
- Song, M., and D. Civco. 2004. "Road Extraction Using SVM and Image Segmentation." *Photogrammetric Engineering & Remote Sensing* 70 (12): 1365–1371. doi:[10.14358/PERS.70.12.1365](https://doi.org/10.14358/PERS.70.12.1365).
- Swietojanski, P., A. Ghoshal, and S. Renals. 2014. "Convolutional Neural Networks for Distant Speech Recognition." *IEEE Signal Processing Letters* 21 (9): 1120–1124. doi:[10.1109/LSP.2014.2325781](https://doi.org/10.1109/LSP.2014.2325781).
- Wang, M., and S. Zhang. 2011. "Road Extraction from High-Spatial-Resolution Remotely Sensed Imagery by Combining Multi-Profile Analysis and Extended Snakes Model." *International Journal of Remote Sensing* 32 (21): 6349–6365. doi:[10.1080/01431161.2010.508801](https://doi.org/10.1080/01431161.2010.508801).
- Weng, Q. 2012. "Remote Sensing of Impervious Surfaces in the Urban Areas: Requirements, Methods, and Trends." *Remote Sensing of Environment* 117: 34–49. doi:[10.1016/j.rse.2011.02.030](https://doi.org/10.1016/j.rse.2011.02.030).
- Wiedemann, C., and H. Ebner. 2000. "Automatic Completion and Evaluation of Road Networks." *International Archives of Photogrammetry and Remote Sensing* 33 (B3/2; PART 3): 979–986.
- Xiangyun, H., and V. Tao. 2007. "Automatic Extraction of Main Road Centerlines from High Resolution Satellite Imagery Using Hierarchical Grouping." *Photogrammetric Engineering and Remote Sensing* 73 (9): 1049.
- Zeiler, M., and R. Fergus. 2014. "Visualizing and Understanding Convolutional Networks." In *Computer Vision—ECCV 2014*, Vol. 8689, 818–833. Cham: Springer International Publishing. doi:[10.1007/978-3-319-10590-1_53](https://doi.org/10.1007/978-3-319-10590-1_53).
- Zhang, J., X. Lin, Z. Liu, and J. Shen. 2011. "Semi-Automatic Road Tracking by Template Matching and Distance Transformation in Urban Areas." *International Journal of Remote Sensing* 32 (23): 8331–8347. doi:[10.1080/01431161.2010.540587](https://doi.org/10.1080/01431161.2010.540587).
- Zhang, Q., and I. Couloigner. 2007. "Accurate Centerline Detection and Line Width Estimation of Thick Lines Using the Radon Transform." *IEEE Transactions on Image Processing* 16 (2): 310–316. doi:[10.1109/TIP.2006.887731](https://doi.org/10.1109/TIP.2006.887731).